FINAL TERM EXAMINATION

AI-CR-502: Practical data science with python

DS-CR-501: Programming for data Science

Instructions

This project is mandatory and constitutes 50% of your final grade.

Submit a single PDF report following the structure given below.

Submit your well-commented code (.py or .ipynb) separately.

All datasets must be from credible, real-world sources (Kaggle, UCI, data.gov, etc.)

The dataset must have at least 1,000 rows and 10 features.

Plagiarism or using identical datasets across teams will result in zero marks.

Project Question

You are a data analyst hired by a company or organization to solve a real-world problem using data. Your task is to:

Select a real-world dataset from a credible source.

Formulate a specific, actionable business or social problem that this dataset can help address.

Perform a complete data analysis lifecycle including:

Data preprocessing and cleaning

Exploratory Data Analysis (EDA)

Insight generation and visualization

Deliver a professional project report that answers the following questions in the exact format provided.

Report Structure

Section A: Title Page

Project Title, Your Name, Course Name, Date

Section B: Executive Summary

In 200–250 words, summarize your entire project.

Clearly state: (a) the problem, (b) your key findings, and (c) your most important recommendation.

Section C: Introduction and Problem Definition

C.1 What is the background or context of the problem you are solving?

C.2 What specific questions are you trying to answer with this data?

C.3 Why is this analysis important? Who would benefit from it?

Section D: Data Description and Preprocessing

D.1 Describe your dataset: source, number of rows/columns, and all relevant features (provide a data dictionary).

D.2 What missing values did you find and how did you handle them? Justify your method.

D.3 Did you find duplicate records? How did you address them?

D.4 What features engineering, scaling, normalization, or encoding did you perform? Why?

D.5 Did you detect outliers? How did you handle them and why?

Section E: Exploratory Data Analysis - EDA

E.1 What are the key summary statistics and distributions of your individual variables? (Use histograms, box plots, etc.)

E.2 What relationships exist between variables? (Use correlation matrices, scatter plots, heatmaps, etc.)

E.3 What initial patterns, trends, or anomalies did you discover? Describe them.

Section F: Insights, Findings, and Visualizations (25 Marks)

F.1 What are your 3 most significant findings from the analysis?

F.2 For each finding, provide a clear, well-labeled visualization and explain the "so what?" why does this matter?

F.3 How do these findings directly answer your problem statement from Section C?

Section G: Conclusion and Recommendations (15 Marks)

G.1 Summarize your most critical insights in 2-3 paragraphs.

G.2 Provide at least 3 concrete, actionable recommendations. For each recommendation, specify:

Who should act on it?

What exactly should they do?

How does your data support this recommendation?

Section H: Appendix (Optional)

Attach your full Python/R code.

Include any additional visualizations or supplementary material.

Dataset Sources You May Use: Clearly Specify the data source and its credibility

Marking Criteria - How You Will Be Evaluated

Component	Marks
Problem Definition & Data Description	10
Data Preprocessing	10
Exploratory Data Analysis (EDA)	10
Insights & Visualizations	10
Conclusion & Recommendations	5
Executive Summary	5
Total	50